

MUHAMMAD HASEEB BUTT

Data Scientist | Machine Learning Engineer | Computer Vision Researcher

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PROFESSIONAL SUMMARY

Motivated BS Data Science student at GIFT University with hands-on experience in machine learning, deep learning, computer vision, and business intelligence. Demonstrated ability to design and implement predictive models, interactive dashboards, and data pipelines using Python, SQL, Power BI, Tableau, and TensorFlow. Currently contributing as a collaborative researcher on an academic Final Year Project exploring open-set object detection using state-of-the-art foundation models. Adept at translating complex datasets into actionable business insights and research findings. Seeking internship and entry-level roles in data science, machine learning, or AI engineering.

CORE COMPETENCIES

Languages: Python, SQL, Java (academic), PHP (academic)

Machine Learning: Scikit-learn, Linear/Logistic Regression, Ensemble Methods, Fuzzy Integral Methods

Deep Learning: TensorFlow, Keras, CNN Architectures

Computer Vision: OpenCV, Image Segmentation, Object Detection, Medical Imaging

Data Analytics: Pandas, NumPy, Statistical Analysis, Predictive Analytics, EDA

Visualization: Matplotlib, Seaborn, Plotly

Business Intelligence: Power BI, Tableau, Apache Superset, Looker Studio

Databases: MySQL, SQL

Version Control: Git, GitHub

Operating Systems: Windows, Linux (basic)

Research: Open-Set Object Detection, Foundation Models, Academic Paper Writing, IEEE Formatting

EDUCATION

BS Data Science — GIFT University, Gujranwala

2022 – Expected 2026

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Statistics, Data Mining, Business Intelligence

FSc Pre-Engineering — Punjab Group of Colleges (PGC)

2020 – 2022

Score: 748/1100 | Grade: B+

Matric in Science — Sialkot Grammar School

2018 – 2020

Score: 880/1100 | Grade: A

TECHNICAL EXPERIENCE

Data Scientist & Data Analyst (Freelance / Academic Practice)

Dec 2022 – Present

- Analyzed large-scale datasets using Python (Pandas, NumPy) and SQL to extract actionable business insights and support data-driven decision-making.
- Designed and deployed interactive dashboards in Power BI, Tableau, Apache Superset, and Looker Studio for real-time reporting and executive decision support.
- Automated end-to-end reporting pipelines to reduce manual data preparation effort and improve reporting accuracy.
- Collaborated with cross-functional academic and project teams to optimize data workflows and deliver measurable analytical outcomes.
- Applied machine learning models (Scikit-learn) for predictive analytics tasks including regression and classification.

RESEARCH EXPERIENCE

Collaborative Research — Open-Set / Novel Object Detection (Final Year Project)

2025 – Present

Department of Data Science, GIFT University | Supervisor: Sir Usman Ali

- Contributing as a member of a collaborative academic research team investigating open-vocabulary and open-set object detection using state-of-the-art foundation models.
- Conducting comparative analysis of detection frameworks including CLIP-based open-vocabulary detectors and modern zero-shot vision models to benchmark performance on novel object categories.
- Reviewing and synthesizing current literature (2023–2025) on open-world detection, zero-shot learning, and vision-language alignment.
- Supporting the preparation of an IEEE-formatted conference paper documenting methodology, experiments, and findings.

PROJECTS

COVID-19 Detection via Fuzzy Integral CNN Ensemble | *Python, TensorFlow, Keras, OpenCV, NumPy*

- Engineered a deep learning ensemble pipeline combining multiple CNN architectures through fuzzy integral fusion for medical image-based COVID-19 detection.
- Achieved 98.93% classification accuracy on chest X-ray data, demonstrating significant improvement over single-model baselines.
- Implemented custom fuzzy integral aggregation logic to optimally combine confidence scores across ensemble members, improving robustness to individual model errors.

Brain Tumor Segmentation Dashboard | *Python, TensorFlow, OpenCV, Power BI / Tableau*

- Developed a deep learning pipeline for automated brain tumor segmentation from MRI scans, integrating medical image preprocessing and CNN-based pixel classification.
- Built an interactive visualization dashboard to present segmentation results and model performance metrics, supporting clinical decision-making workflows.
- Applied advanced image augmentation techniques to address class imbalance in medical imaging datasets.

House Price Prediction — Linear Regression | *Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn*

- Implemented a Linear Regression model on the Kaggle House Prices dataset to predict residential property values based on square footage, bedrooms, and bathrooms. [GitHub: github.com/Haseebbutt89/PRODIGY_MAY25_TaskNumber_1](https://github.com/Haseebbutt89/PRODIGY_MAY25_TaskNumber_1)
- Conducted end-to-end ML workflow: data cleaning, feature selection, model training, and performance evaluation using MSE and R^2 metrics.
- Delivered clear performance visualizations to communicate model accuracy and prediction residuals.

AutoStatAgent — Automated Statistical Analysis & Report Generation | *Python, Pandas, NumPy, Matplotlib, LaTeX*

- Built an automated data analysis tool that ingests structured datasets, performs statistical analysis, and generates formatted LaTeX PDF reports without manual intervention.
- Reduced report generation time by automating descriptive statistics, distribution analysis, and visualization rendering.
- Designed modular architecture enabling plug-and-play support for diverse dataset types.

Business Intelligence Dashboards — Multi-Domain Analytics | *Power BI, Tableau, Apache Superset, Looker Studio, SQL*

- Designed and delivered multiple BI dashboards for academic and practice projects covering sales analytics, performance tracking, and operational metrics.
- Implemented dynamic filtering, drill-down capabilities, and KPI cards to enable real-time self-service analytics.
- Utilized Apache Superset and Looker Studio as open-source and cloud-based alternatives to enterprise BI platforms.

CERTIFICATIONS

- Applied Data Science with Python — IBM / Coursera | 2025
- Professional Business Analysis — IBM / Coursera | 2025

ACHIEVEMENTS

- Achieved 98.93% accuracy in COVID-19 chest X-ray classification using a novel fuzzy integral CNN ensemble approach.
- Selected as a contributor in a GIFT University research team targeting IEEE conference publication on open-set object detection.
- Completed two IBM-certified professional programs (Data Science with Python & Business Analysis) in 2025.
- Prodigy InfoTech Machine Learning Track participant — completed supervised learning task on Kaggle competition dataset.

LANGUAGES

- Urdu — Native

- English — Professional Working Proficiency

INTERESTS & AREAS OF FOCUS

Data Science | Machine Learning | Deep Learning | Computer Vision | Business Intelligence | Predictive Analytics |
Research & Academic Publishing | Medical Imaging AI